

# ITS - Industrial Technology Solutions

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## CNC Plasma Cutting Machine - Service Report

|                   |                                           |
|-------------------|-------------------------------------------|
| <b>Client:</b>    | SOMIK Skikda - Skikda                     |
| <b>Date:</b>      | 22/07/2026                                |
| <b>Machine:</b>   | KOTEC KTPG-320 CNC Plasma Cutting Machine |
| <b>Serial No:</b> | 149P324050WYPV1K                          |

### Machine Specifications

| Parameter           | Details                                 |
|---------------------|-----------------------------------------|
| Manufacturer        | KumKwang Tech Co., Ltd. (KOTEC) - Korea |
| Model               | KTPG - 320                              |
| Date of Manufacture | November 2010                           |
| Controller          | Sentrol 2 (SENTROL)                     |
| Servo Motors        | YE LI YBL13S-75L.Z (1kW, 2000 RPM)      |
| Servo Drives        | YPV Series AC Servo Drives              |
| Plasma System       | Hypertherm HyPerformance Plasma         |

### Part 1: Movements Control

Invoice No: ITS-2026-001

**Problem:** The machine has no PLC (Programmable Logic Controller). The Sentrol 2 CNC controller relies on electromagnetic relays for I/O, limiting motion control capabilities.

#### Work Performed:

1. Installed Mach3 software on dedicated PC All-in-One
2. Configured YE LI YBL13S-75L.Z servo motors (X, Y, Z axes)
3. Set up YPV Series servo drives
4. Adapted Mach3 with servo drives
5. Tested the first motor

6. Rewired all schematic control for the new system
7. Customized Mach3 interface
8. Configured travel limits and home positions
9. Tested full system
10. Movement tests and operation validation

**Parts / Materials Used:**

| No. | Description                        | Qty | Local Price (DA) | Import Price (EUR) | Total         |
|-----|------------------------------------|-----|------------------|--------------------|---------------|
| 1   | PC All-in-One                      | 1   | 40,000           | -                  | 40,000 DA     |
| 2   | Mach3 License                      | 1   | -                | 165 - 189          | 165 - 189 EUR |
| 3   | Mach3 USB Interface Board (4-axis) | 1   | 5,400            | 26 - 37            | 5,400 DA      |
| 4   | Box Panel                          | 1   | 5,000            | -                  | 5,000 DA      |

**Part 1 Total: 400,000.00 DA**

**Part 2: Plasma Out**

**Invoice No: ITS-2026-002**

**Problem:** The plasma system components exist at the machine but need to be integrated and configured with the new Mach3 CNC control system.

**Work Performed:**

1. Integrated Hypertherm HyPerformance Plasma system with Mach3 CNC
2. Configured Ignition Console (078172)
3. Configured Manual Gas Mix Console (078532)
4. Set up Gas Control and Metering Panel (O2/Air/N2)
5. Connected plasma torch and torch leads
6. Configured safety interlocks (coolant, pressure, voltage)

## 7. Cutting tests and operation validation

### Components Already at Machine:

| No. | Component                               | Part No. | Status  |
|-----|-----------------------------------------|----------|---------|
| 1   | HyPerformance Plasma Power Supply (HPR) | -        | We Have |
| 2   | Ignition Console                        | 078172   | We Have |
| 3   | Manual Gas Mix Console                  | 078532   | We Have |
| 4   | Plasma Torch and Torch Leads            | -        | We Have |
| 5   | Gas Control and Metering Panel          | -        | We Have |
| 6   | Work Cable (Grounding)                  | -        | We Have |

### Parts / Materials Used:

| No. | Description | Qty | Local Price (DA) | Import Price (EUR) | Total |
|-----|-------------|-----|------------------|--------------------|-------|
| 1   |             |     |                  |                    |       |
| 2   |             |     |                  |                    |       |
| 3   |             |     |                  |                    |       |
| 4   |             |     |                  |                    |       |
| 5   |             |     |                  |                    |       |

**Part 2 Total: 150,000.00 DA**

### Summary

| Invoice      | Service           | Amount        |
|--------------|-------------------|---------------|
| ITS-2026-001 | Movements Control | 400,000.00 DA |
| ITS-2026-002 | Plasma Out        | 150,000.00 DA |

**TOTAL: 550,000.00 DA**

**For ITS**

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Signature and Stamp